Hamza Rashid

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EDUCATION

University of Georgia, College of Engineering – Athens, GA — Aug. 2026 – Dec. 2029

Bachelor of Science in Electrical and Electronics Engineering

Relevant Coursework: Physics, Intro to Electrical & Computer Systems Engineering, Biometric Systems

TECHNICAL PROFICIENCIES

Hardware: Circuit Design & Analysis, Soldering, Breadboarding, PCB Layout & Schematic Capture, Sensors (I2C, PWM, Time-of-Flight), Digital Multimeters (DMM), Oscilloscopes, Bench Power Supplies, Raspberry Pi, Arduino, Microcontrollers
Software: Python, HTML/CSS, Multisim, VEXcode, Visual Studio Code, Autodesk Fusion 360, OpenCV, DeepFace, Flask, GPIO/ADC/SPI/UART Peripheral Interfacing, Cisco Packet Tracer (Network Simulation), Wireshark (Live Traffic Analysis)
Languages & Certifications: English (Fluent), Bengali (Fluent), Spanish (Limited Working Proficiency)
Google AI Professional Certificate (Google, Oct. 2026), Cyber Security: Beginner to Expert (Udemy, 2023)

PROJECT EXPERIENCE

AgriGuard: Autonomous Pest-Deterrent System, IEEE FYC Team Project – Athens, GA — Sept. 2026 – Present

- Collaborating in a team of **4**, developing a Raspberry Pi and OpenCV vision pipeline (**15 FPS**, targeting **85%** detection precision under controlled lighting) that drives **2** pan-and-tilt servos via an I2C PWM controller to align the spray nozzle within **5 degrees** of each detected target, keeping aim accurate and repeatable across the camera's entire field of view
- Using a nozzle-mounted time-of-flight sensor to range targets from **0.5 to 1.5 meters** (targeting distance error under **5 cm**) before switching a GPIO-controlled relay to fire the DC water pump in approximately **500 ms** bursts, targeting successful hits in **70 of 100** trials across repeated test runs, aiming to protect agriculture from pests through deterrence

Credit Scoring Platform for Egypt, KSU Fintech Hackathon – Kennesaw, GA — Mar. 2026

- Placed **2nd of 40 teams** at a hackathon building a proof-of-concept fintech platform (Python, Flask, Pandas) pairing e-wallet data with secure government ID verification to assess credit access for historically underbanked users in Egypt
- Integrated the live Egypt National ID API (Arya.ai) for identity verification and OpenCV with DeepFace for facial matching against the ID photo; credit-scoring itself was a theoretical, rules-based demo, with encryption and KYC compliance scoped as required hardening steps for a full real-world production deployment beyond the scope of this initial prototype

VEX Disc-Golf Launch Robot, VEX Robotics, Technology Student Association – Martinez, GA — Fall 2022

- Designed and wired the robot's flywheel launcher subsystem, driving **2** 393 motors through VEX Motor Controller 29s off a **7.2V** NiMH battery pack to generate consistent long-range disc launch velocity, ensuring every launch lands accurately
- Programmed flywheel speed control in VEXcode using a quadrature shaft encoder for closed-loop RPM feedback, reducing shot-distance variance and placing **2nd in state** against highly competitive teams from across the entire state of Georgia

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant, UGA Cultivate Lab – Athens, GA — Oct. 2026 – Present

- Shaping curriculum design for ECSE 1100 (Intro to Electrical & Computer Systems) under Dr. Sarah Jane Bork, impacting **350+** students annually by integrating AI and, with **2** researchers, redesigning **3** lectures, **2** labs, and assessments

SAT Tutor, Independent / Schoolhouse.world (Khan Academy) – Remote — 2025 – Present

- Tutored students in SAT prep, breaking down reasoning strategies; one student's score rose from **1100 to 1300** in a month using data-driven, personalized study plans and analytical, standardized-testing techniques for each learner's own needs

Groundskeeping Crew, The Groundsman – Augusta, GA — Apr. 2026

- Operated golf carts and transported equipment for The Masters Tournament, working fast-paced early-morning shifts to keep logistics running for daily crowds under tight time constraints, ensuring smooth operations for thousands of people

LEADERSHIP & AWARDS

CyberPatriot, Team Captain, Lakeside High & Stallings Island Middle School – Martinez, GA — 2020 – 2026

- Achieved Platinum Tier all **six** years in the Air Force Association's CyberPatriot competition, ranking **top 30%** at State by securing Windows/Linux systems and analyzing network traffic with Wireshark in timed, multi-round team competitions